

Assessment 1 : Project 3 - Aircraft Classification - Report

This project investigated transfer-learning approaches for classifying images from the 20-class FGVC-Aircraft subset. After inspection and data splitting (part 1) a baseline model was established (part 2) and three controlled experiments were conducted (part 3) using the same training/validation split, with model-development decisions based only on validation performance.

Table 1: Validation Performance across controlled experiment

Run	Controlled Change	Best Val Acc	Best Epoch
Baseline	[baseline configuration]	0.805243	10
Experiment 1	Data Augmentation	0.850187	14
Experiment 2	Learning-rate tuning	0.835206	15
Experiment 3	EfficientNet-B0 backbone	0.794007	14

Figure 1: Validation Accuracy Learning Curves

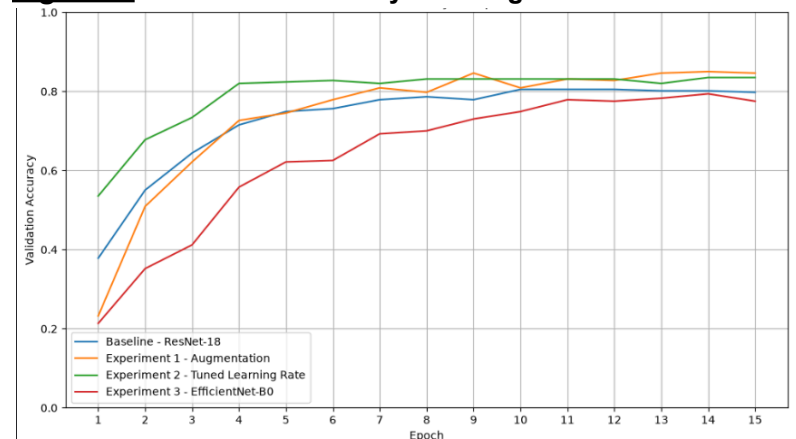


Figure 2: Per Class test accuracy - winning model

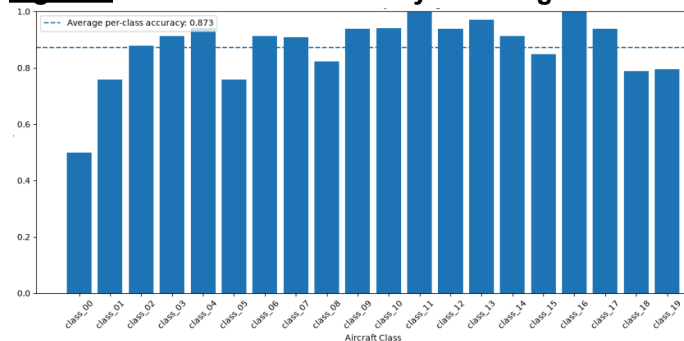


Figure 3: Winning Model Best and Worst Test Accuracy

Highest-performing classes:			Lowest-performing classes:		
Aircraft Class	Test Accuracy		Aircraft Class	Test Accuracy	
16	class_16	1.000000	1	class_01	0.757576
11	class_11	1.000000	5	class_05	0.757576
13	class_13	0.969697	0	class_00	0.500000

Table 1 summarises validation performance across the baseline and controlled experiments, while **Figure 1** compares their learning curves. The final winning model achieved an average per-class test accuracy of **0.873**, per class test accuracy can be seen in **Figure 2** and **Figure 3**.

Baseline

Baseline preprocessing consisted of converting images to tensors, resizing them to 224×224 pixels, and normalising using ImageNet mean and standard deviation values. No data augmentation was applied.

ResNet-18 was used as the baseline with unfrozen backbone, the aforementioned preprocessing, **batch size 16**, **learning rate 0.001**, **SGD with momentum 0.9**, and **15 epochs**. This provided a simple transfer-learning starting point against which subsequent controlled experiments could be evaluated.

The baseline achieved a best-validation-accuracy of **0.805** at epoch **10** (ref **Table 1**). Its learning curve (**Blue**) in **Figure 1** indicates rapid improvement over the early epochs followed by a plateau at approximately 0.80 from epoch 10 onward, suggesting that the baseline had largely converged within the 15-epoch training period

Controlled Experiment 1

Hypothesis: *Introducing random image augmentation may improve generalisation by exposing the model*

to greater variation in aircraft orientation and appearance while preserving the class label.

The baseline training transform was modified by adding **RandomHorizontalFlip**, **RandomRotation(15)** and **ColorJitter** with **brightness/contrast/saturation=0.2** and **hue=0.1**, while the architecture, optimiser, learning rate, batch size, and training duration were kept unchanged. This experiment achieved a best-validation-accuracy of **0.850**, representing a +0.045 change relative to baseline (ref **Table 1**).

The result **supports the hypothesis** because although augmentation initially slowed validation performance, the model overtook the baseline during later epochs and ultimately achieved the highest validation accuracy of the three controlled experiments, suggesting improved generalisation from the additional training-image variation.

Controlled Experiment 2

Hypothesis: *The baseline learning rate may not provide the most effective optimisation behaviour, so tuning it could improve validation performance without changing the model architecture.*

Optuna was used to evaluate 10 candidate learning rates while keeping the remaining training configuration fixed. The optimal learning rate was found to be 0.004089, which produced a best-validation-accuracy of **0.835** at epoch **15** (ref **Table 1**).

Compared with the baseline, this was an improvement of +0.030. **Figure 1** shows that the tuned model learned substantially faster than the baseline, reaching approximately 0.82 validation accuracy by epoch 4 and then remaining relatively stable. This suggests that the larger tuned learning rate improved optimisation speed and produced a modest improvement in final validation performance.

Controlled Experiment 3

Hypothesis: *A different pretrained CNN architecture may provide feature representations better suited to fine-grained aircraft classification.*

The **ResNet-18** backbone was replaced with **EfficientNet-B0**, while the remaining controlled settings were kept as comparable as possible. EfficientNet achieved a best-validation-accuracy of **0.794**, compared with **0.805** for the baseline.

This reduced validation performance by 0.011. Although EfficientNet improved steadily throughout training and approached the baseline by the later epochs, it never surpassed it, suggesting that changing to the more modern architecture alone did not improve generalisation under the selected training configuration.

Best Model

Based on validation performance, the selected final configuration was **Experiment 1**. It achieved the strongest validation accuracy of **0.850**, compared with **0.835** for the next best, Experiment 2. The model was selected because it achieved the highest validation accuracy while avoiding the additional computational cost of Optuna tuning.

The winning configuration was then retained from a fresh pretraining ResNet-18 using the complete **trainval** dataset and evaluated once on the held-out test set. It achieved an **average per-class accuracy of 0.873**, exceeding the task's reference performance of 0.75. Per class performance ranged from **0.500** for class_00 to **1.00** for class_16 and class_11 (Ref **Figure 3**).

The result demonstrates the potential of pretrained CNNs and transfer learning for automated fine-grained aircraft recognition. However, the substantial variation between classes indicates that some aircraft remain considerably harder to distinguish (ref **Figure 2**). The relatively small curated dataset may also limit generalisation to unseen viewpoints, lighting, background noise. For this, further improvement could investigate targeted augmentation, additional training data, or class-specific error analysis.